

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 10 · Trigonometric Equations

Questions

Student book · questions with writing space

- 46 pages
- Past papers up to January 2026

PURE 2 | TOPIC 10

Trigonometric Equations

Your classified route

20

QUESTIONS

173

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 10 | FAMILY 01

Trig Equations

WHAT THIS FAMILY ASKS

Recognise the trig equations demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**9 classified questions**

79 total marks

9. Solutions based entirely on graphical or numerical methods are not acceptable in this question.

(i) Solve, for $0 \leq \theta < 180^\circ$, the equation

$$3 \sin(2\theta - 10^\circ) = 1$$

giving your answers to one decimal place.

(4)

Adaptive unit 1 · WMA12-01_2019_Oct_Q09

DR ESAM AHMED

9. Solutions based entirely on graphical or numerical methods are not acceptable in this question.

(ii) The first three terms of an arithmetic sequence are

$$\sin \alpha, \frac{1}{\tan \alpha} \text{ and } 2 \sin \alpha$$

where α is a constant.

(a) Show that $2 \cos \alpha = 3 \sin^2 \alpha$

(3)

Given that $\pi < \alpha < 2\pi$,

(b) find, showing all working, the value of α to 3 decimal places.

(5)

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6. (a) Show that the equation

$$\frac{3 \sin \theta \cos \theta}{2 \sin \theta - 1} = 5 \tan \theta \quad \sin \theta \neq \frac{1}{2}$$

can be written in the form

$$3 \sin^3 \theta + 10 \sin^2 \theta - 8 \sin \theta = 0$$

(4)

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(b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3\sin 2x \cos 2x}{2\sin 2x - 1} = 5 \tan 2x$$

giving your answers to 3 decimal places where appropriate.

(4)

Adaptive unit 2 · WMA12-01_2021_Jan_Q06

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7. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $-90^\circ < x < 90^\circ$, the equation

$$3 \sin(2x - 15^\circ) = \cos(2x - 15^\circ)$$

giving your answers to one decimal place.

(4)

Adaptive unit 1 · WMA12-01_2022_Jan_Q07

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7. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

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5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Solve, for $-180^\circ < \theta \leq 180^\circ$, the equation

$$3 \tan(\theta + 43^\circ) = 2 \cos(\theta + 43^\circ)$$

(6)

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3.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(a) Solve, for $0 < \theta \leq 360^\circ$ the equation

$$2 \tan \theta + 3 \sin \theta = 0$$

giving your answers, as appropriate, to one decimal place.

(5)

Adaptive unit 1 · WMA12-01_2023_Oct_Q03

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3. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(b) Hence, or otherwise, find the smallest positive solution of

$$2 \tan (2x + 40^\circ) + 3 \sin (2x + 40^\circ) = 0$$

giving your answer to one decimal place.

(2)

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(6)

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

(ii)

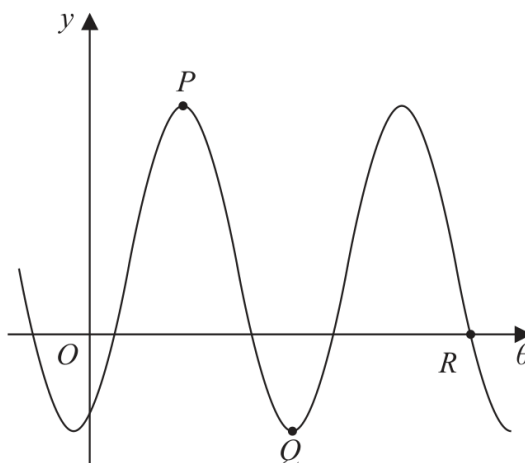
**Figure 1**

Figure 1 shows a sketch of part of the curve with equation

$$y = A \sin\left(2\theta - \frac{3\pi}{8}\right) + 2$$

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where A is a constant and θ is measured in radians.

The points P , Q and R lie on the curve and are shown in Figure 1.

Given that the y coordinate of P is 7

- (a) state the value of A , (1)
- (b) find the exact coordinates of Q , (3)
- (c) find the value of θ at R , giving your answer to 3 significant figures. (4)

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7. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

(i) Solve, for $0 \leq x < 2\pi$, the equation

$$3 \sin x \tan x = 11 + \cos x$$

giving the answers in radians to 3 decimal places.

(5)

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7. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Given that

- $0 < \theta < 90^\circ$
- $\cos \theta = \frac{1}{3}$

find, in simplest form, the exact value of $\tan \theta$

(2)

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8:

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

(i) Solve, for $0 \leq x < \pi$, the equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3}$$

giving the answers in radians in the form $k\pi$, where k is a rational constant to be found.

(3)

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8: In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta \tan \theta = \cos \theta + 4$$

giving your answers, in degrees, to one decimal place.

(5)

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5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < \theta \leq 360^\circ$, the equation

$$4 \tan \theta + 5 \sin \theta = 0$$

giving any non-exact answers to one decimal place.

(5)

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5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Solve, for $0 < x < \pi$, the equation

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{5}{\cos x}$$

giving the answers, in radians, to 3 significant figures.

(4)

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TOPIC 10 | FAMILY 02

Trig Identities

WHAT THIS FAMILY ASKS

Recognise the trig identities demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

2 classified questions

15 total marks

7. (a) Show that the equation

$$8 \tan \theta = 3 \cos \theta$$

may be rewritten in the form

$$3 \sin^2 \theta + 8 \sin \theta - 3 = 0$$

(3)

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(b) Hence solve, for $0 \leq x \leq 90^\circ$, the equation

$$8 \tan 2x = 3 \cos 2x$$

giving your answers to 2 decimal places.

(4)

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10. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) \equiv \sin \theta + \frac{2}{\sin \theta} \qquad \theta \neq \frac{n\pi}{2}$$

(4)

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10. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$\cos x \left(3 \tan x + \frac{2}{\tan x} \right) = 4 \sin x - 5$$

giving your answers to 3 significant figures.

(4)

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TOPIC 10 | FAMILY 03

Periodicity

WHAT THIS FAMILY ASKS

Recognise the periodicity demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

10 total marks

10. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3$$

(5)

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10. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Solve, for $0 < \theta < 360^\circ$

$$(2 \sin \theta - \cos \theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place.

(5)

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TOPIC 10 | FAMILY 04

Transformed Trig

WHAT THIS FAMILY ASKS

Recognise the transformed trig demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 Expand the interval for the inside expression before finding angles.

8 classified questions

69 total marks

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < \theta < 360^\circ$, the equation

$$3 \sin(\theta + 30^\circ) = 7 \cos(\theta + 30^\circ)$$

giving your answers to one decimal place.

(4)

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8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x (a \cos^2 x + b \cos x + c) = 0$$

where a , b and c are constants to be found.

(b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

(6)

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8. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

(i) Solve, for $-\frac{\pi}{2} < x < \pi$, the equation

$$5 \sin(3x + 0.1) + 2 = 0$$

giving your answers, in radians, to 2 decimal places.

(4)

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8. In this question you must show all stages of your working.
Solutions based entirely on calculator technology are not acceptable.

(ii) Solve, for $0 < \theta < 360^\circ$, the equation

$$2 \tan \theta \sin \theta = 5 + \cos \theta$$

giving your answers, **in degrees**, to one decimal place.

(5)

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8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

(i) Solve, for $0 < x \leq \pi$, the equation

$$5 \sin x \tan x + 13 = \cos x$$

giving your answer in radians to 3 significant figures.

(5)

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8.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) The temperature inside a greenhouse is monitored on one particular day.

The temperature, $H^{\circ}\text{C}$, inside the greenhouse, t hours after midnight, is modelled by the equation

$$H = 10 + 12 \sin(kt + 18)^{\circ} \quad 0 \leq t < 24$$

where k is a constant.

Use the equation of the model to answer parts (a) to (c).

Given that

- the temperature inside the greenhouse was 20°C at 6 am
- $0 < k < 20$

(a) find all possible values for k , giving each answer to 2 decimal places.

(4)

Given further that $0 < k < 10$

(b) find the maximum temperature inside the greenhouse,

(1)

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(c) find the time of day at which this maximum temperature occurs.
Give your answer to the nearest minute.

(2)

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5.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

(a) Show that the depth of water in the harbour at 2 am is just over 4 metres.

(1)

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5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

(b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres.

(5)

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9. **In this question you must show detailed reasoning.**
Solutions relying entirely on calculator technology are not acceptable.
(a) Show that the equation

$$2 \tan \theta = 3 \cos \theta$$

can be written as

$$3 \sin^2 \theta + 2 \sin \theta - 3 = 0$$

(3)

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9. In this question you must show detailed reasoning.
Solutions relying entirely on calculator technology are not acceptable.

(b) Hence solve, for $-\pi < x < \pi$, the equation

$$2 \tan\left(2x + \frac{\pi}{3}\right) = 3 \cos\left(2x + \frac{\pi}{3}\right)$$

giving your answers to 3 significant figures.

(4)

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5.

In this question you must show detailed reasoning.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that the equation

$$(3 \cos \theta - \tan \theta) \cos \theta = 2$$

can be written as

$$3 \sin^2 \theta + \sin \theta - 1 = 0$$

(3)

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5. In this question you must show detailed reasoning.
Solutions relying entirely on calculator technology are not acceptable.

(b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$(3 \cos 2x - \tan 2x) \cos 2x = 2$$

(5)

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9.

In this question you must show all stages of your working.**Solutions relying entirely on calculator technology are not acceptable.**

(a) Show that

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 11 - 5 \cos \theta$$

may be written as

$$3 \cos^2 \theta - 14 \cos \theta + 8 = 0$$

(3)

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9. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(b) Hence solve, for $0 < x < 360^\circ$

$$3 \cos 2x (\tan 2x \sin 2x + 3) = 11 - 5 \cos 2x$$

giving your answers to one decimal place.

(4)

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8. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

(i) Solve, for $0 \leq \theta < 180$

$$4 \tan^2 (2\theta - 30)^\circ + 1 = 49$$

giving the answers to one decimal place.

(5)

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8. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Solve, for $0 \leq x < 2\pi$

$$2 \tan x \sin x + 3 = 0$$

giving the answers, in radians, in the form $k\pi$ where k is a rational constant.

(5)

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